Know what to check before you stack a single block.

Field-tested across 5 phases · Honest · Built for first-time ICF builders

25 CHECKPOINTS

1 DESIGN REVIEW

- [ ] **Plans are ICF-native** — CRITICAL
  Not converted from a wood frame plan. ICF walls have different corner geometry, lintel depths, and floor tie-in details — converting a wood plan introduces structural gaps.
- [ ] **Core thickness specified**
  6" core standard for above-grade residential walls. 8" for below-grade or two-story. Confirm with your engineer before ordering block.
- [ ] **Window & door RO schedule from manufacturer** — CRITICAL
  Rough opening dimensions must come from the window/door manufacturer — not assumed from nominal size. Get this before building a single buck.
- [ ] **HVAC sized for ICF airtightness**
  ICF homes run 1.0–1.75 ACH50. An oversized unit short-cycles and can't dehumidify. "Google 'ICF and rainy walls' — that's an oversized AC." Downsize vs. wood frame.
- [ ] **Mechanical ventilation planned**
  ERV or HRV required. ICF walls are too airtight for passive exchange. Plan sleeve locations and intake points at the design phase — impossible to add cleanly after the walls are poured.

2 PERMIT & ENGINEERING

- [ ] **CCRR-1003 obtained for building department**
  The ICC code compliance research report. Building inspectors who haven't seen ICF will ask for it. Have it in your permit package before the first meeting.
- [ ] **Rebar schedule specified by engineer** — CRITICAL
  PSI, bar size, and spacing per ACI 318. Typical residential: #4 bar at 24–36" OC vertical; every other course horizontal above grade. Heavy clay soil requires tighter spacing.
- [ ] **Engineer has ICF experience**
  Wood-frame engineers sometimes over-specify ICF rebar (as tight as 6" OC on everything). An ICF-experienced engineer designs to actual slender wall tables, not conservative guesses.
- [ ] **Anchor bolt layout on drawings**
  Anchor bolts must be set within 20–30 minutes of the top of pour. Layout must be marked on the wall before the truck arrives — no time to tape-measure while concrete sets.
- [ ] **Energy compliance documentation ready**
  HERS rating, blower door test requirement, and energy code path (prescriptive or performance) confirmed with your jurisdiction before permit submission.

3 SITE & FOUNDATION

- [ ] **Benchmark elevation set** — CRITICAL
  Single fixed benchmark (nail in tree or permanent stake) at finished floor elevation. Every trade references the same point through the entire project. No benchmark = drifting floor heights.
- [ ] **Step footings in 16" or 8" increments only**
  Never 12". A 12" step cuts through the middle of a web tie, requiring custom cuts on every block at that transition. 16" = one full block height. 8" = half block.
- [ ] **Footing poured level within ¼"**
  Level footings make the wall phase fast. Out-of-level footings mean shimming every course — the block layer never gets into a rhythm. Crown footing boards face down for ICF.
- [ ] **Vertical dowels at correct spacing**
  Set per engineering tables. Spray-paint them bright orange — inspector verifies lap from the scaffold. "Their inspection for us is a lintel inspection." Orange paint = zero re-inspection delays.
- [ ] **Bracing confirmed: quantity, rental, delivery date**
  Rule of thumb: (linear footage ÷ 6) + number of corners = brace count. A typical house needs 32–48 braces. Confirm rental availability and delivery to site before stacking begins.

4 STACK & POUR

- [ ] **Concrete ordered to spec** — CRITICAL
  4,000 PSI · 3/8" aggregate · 6–6.5" slump · water reducer. Specify "ICF pour" to the plant — not a flatwork mix. Confirm they stock 3/8" aggregate before pour day.
- [ ] **Pump truck scheduled; hose type confirmed**
  4" flat pressure-actuated hose preferred over 3" with double-S (double-S being phased out in many states for safety). Confirm hose type with the pump operator before pour day.
- [ ] **Bucks installed, plumb, braced, and check-holed** — CRITICAL
  Every window and door buck: set to RO spec, plumb and square, cross-braced every 2 feet, check holes drilled in sill bottom for vibrator access. Confirm nailer orientation before stacking above.
- [ ] **All penetrations roughed in before pour**
  Plumbing, electrical, and HVAC sleeves must be placed before the walls are poured. There is no going back after concrete sets. Walk the wall with each sub the day before.
- [ ] **Pour crew assigned: 5 people minimum**
  1 on hose · 1 on vibrator · 1 each side of wall monitoring fill level · 1 utility/patch. Anchor bolt layout marked on wall top before truck arrives. String lines tensioned for active plumb-check during pour.

5 POST-POUR

- [ ] **Anchor bolts set in window** CRITICAL
  Set within 20–30 min of top of pour. Use sill seal (neoprene) between concrete and sill plate — never lumber directly on concrete.

- [ ] **Wall checks at 6hr and 24hr**
  Check plumb, check for voids at top of pour, check for any residual bulge at window/door heads. Correct at 6 hours — not at 24.

- [ ] **7-day cure before backfill**
  Minimum 7 days at temps above 50°F for a 3,500 PSI mix. Extend in cold weather. Do not backfill until floor system is in place per ACI 318 slender wall provisions.

- [ ] **Window flashing in sequence** CRITICAL
  Sill first → sides → head. Then set window, then cover fin. Never head before jambs. Leave sill drainage gap — do not fully seal the bottom.

- [ ] **HVAC sub briefed on ICF sizing**
  Confirm tonnage is right-sized for ICF. Variable-speed air handler required for proper dehumidification. Mechanical ventilation (ERV/HRV) commissioned before occupancy.

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